Applied Category Theory

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Option A

Task 1

1. Resource preorders

We order resources according to usefulness: $x \leq y$ means that y can do everything that x can do in the relevant interface.

1. $\mathsf{Altitude} = (\mathbb{R}_{\geq 0}, \leq)$

   A higher available altitude satisfies any lower minimum-altitude requirement.

2. $\mathsf{RadLevel} = \{\text{Low} \leq \text{Medium} \leq \text{High}\}$

   An orbital slot requires radiation tolerance at least equal to its exposure level, while a bus provides a radiation-tolerance capability.

3. $\mathsf{LaunchWindows} = (\mathcal{P}\{Q_1, Q_2, Q_3\}, \subseteq)$

   A larger set of available launch windows is interpreted as more useful.

4. $\mathsf{MassBudget} = (\mathbb{R}_{\geq 0}, \leq)$

   Here an element denotes available mass capacity. A design whose physical mass is m requires a mass budget of at least m.

5. $\mathsf{Power} = (\mathbb{R}_{\geq 0}, \leq)$

   An amount p of consumed power requires an available power budget of at least p. The bus's Power column denotes payload-support power capacity.

6. $\mathsf{DataRate} = (\mathbb{R}_{\geq 0}, \leq)$

   The payload produces a required data rate, while the bus provides downlink capacity.

7. $\mathsf{ScienceValue} = (\mathbb{R}_{\geq 0}, \leq)$

8. $\mathsf{DebrisRisk} = (\mathbb{R}_{\geq 0}, \leq_{\text{risk}})$, where $r \leq_{\text{risk}} r' \iff r \geq r'$ numerically

   Thus a smaller numerical risk is a more useful resource.

2. Bool-enriched profunctor model

Following the co-design convention, a subsystem with provided-resource preorder P and required-resource preorder R is represented by a Bool-enriched profunctor

$$\Phi : P \nrightarrow R, \qquad \Phi : P^{\text{op}} \times R \longrightarrow \mathbf{Bool}.$$

The statement $\Phi(p, r) = \text{true}$ means that the subsystem can provide functionality p when supplied with resources r.

The implementation sets $D_{\text{slot}},\ D_{\text{bus}},\ D_{\text{pay}}$ index the possible implementations. For each particular implementation, we define a separate profunctor.

The subsystem interfaces are:

Subsystem	Resources provided	Resources required
Orbital slot	Altitude LaunchWindows DebrisRisk	RadLevel
Satellite bus	Power DataRate RadLevel	Altitude MassBudget
Payload	ScienceValue	Power DataRate MassBudget

Internal coupling

The internally connected resources are

$$\begin{aligned}\text{slot altitude} &\longrightarrow \text{bus altitude requirement},\\ \text{bus radiation tolerance} &\longrightarrow \text{slot radiation-tolerance requirement},\\ \text{bus power capacity} &\longrightarrow \text{payload power requirement},\\ \text{bus downlink capacity} &\longrightarrow \text{payload data-rate requirement}.\end{aligned}$$

Orbital-slot implementations

For each orbital slot $\ell \in D_{\text{slot}}$, write

$$A_\ell, \qquad W_\ell, \qquad E_\ell, \qquad R_\ell$$

for its altitude, launch-window set, radiation exposure, and numerical debris risk.

The implementation ℓ determines the profunctor

$$\Phi^{\ell}_{\text{slot}} : \mathsf{Altitude} \times \mathsf{LaunchWindows} \times \mathsf{DebrisRisk} \nrightarrow \mathsf{RadLevel},$$

defined by

$$\Phi^{\ell}_{\text{slot}}\big((a, W, \rho), \tau\big) = \text{true} \iff a \le A_\ell,\ W \subseteq W_\ell,\ \rho \le_{\text{risk}} R_\ell,\ E_\ell \le \tau.$$

Since $\rho \le_{\text{risk}} R_\ell \iff R_\ell \le_{\mathbb{R}} \rho$, the slot's actual debris risk must not exceed the requested upper bound ρ. Thus a slot provides altitude, launch-window availability, and debris-risk performance, while requiring sufficient radiation tolerance.

Satellite-bus implementations

For each bus $b \in D_{\text{bus}}$, write

$$M_b, \qquad P_b, \qquad D_b, \qquad T_b, \qquad A_b^{\min}$$

for its mass, payload-support power capacity, downlink capacity, radiation tolerance, and minimum operating altitude.

The implementation b determines the profunctor

$$\Phi^{b}_{\text{bus}} : \mathsf{Power} \times \mathsf{DataRate} \times \mathsf{RadLevel} \nrightarrow \mathsf{Altitude} \times \mathsf{MassBudget},$$

defined by

$$\Phi^{b}_{\text{bus}}\big((p, d, \tau), (a, m_b)\big) = \text{true} \iff p \le P_b,\ d \le D_b,\ \tau \le T_b,\ A_b^{\min} \le a,\ M_b \le m_b.$$

Thus the bus provides power capacity, downlink capacity, and radiation tolerance. It requires a sufficiently high orbital altitude and a mass budget large enough to accommodate the bus.

Payload implementations

For each payload $y \in D_{\mathrm{pay}}$, write

$$P_y, \qquad D_y, \qquad M_y, \qquad S_y$$

for its required power, required data rate, mass, and science value.

The implementation y determines the profunctor

$$\Phi^{y}_{\mathrm{pay}} : \mathsf{ScienceValue} \nrightarrow \mathsf{Power} \times \mathsf{DataRate} \times \mathsf{MassBudget},$$

defined by

$$\Phi^{y}_{\mathrm{pay}}\big(s, (p, d, m_y)\big) = \mathrm{true} \iff s \le S_y,\ P_y \le p,\ D_y \le d,\ M_y \le m_y.$$

Thus the payload provides science value and requires sufficient power, data-rate capacity, and mass budget.

Mass aggregation

The bus and payload require separate local mass budgets m_b and m_y. A single external total-mass budget is represented by

$$\Phi_{\mathrm{mass}} : \mathsf{MassBudget} \times \mathsf{MassBudget} \nrightarrow \mathsf{MassBudget},$$

$$\Phi_{\mathrm{mass}}\big((m_b, m_y), m\big) = \mathrm{true} \quad \iff \quad m_b + m_y \le m.$$

Feasibility predicate on implementation triples

Equip $D_{\mathrm{slot}},\ D_{\mathrm{bus}},\ D_{\mathrm{pay}}$ with the discrete preorder. The intrinsic feasibility relation is the Bool-enriched profunctor

$$\chi : D_{\mathrm{slot}} \times D_{\mathrm{bus}} \times D_{\mathrm{pay}} \nrightarrow \mathbf{1},$$

where $\mathbf{1}$ is the one-element preorder.

For a triple (ℓ, b, y), define

$$\chi(\ell, b, y) = \Phi^{\ell,b,y}_{\mathrm{total}}\big((S_y, W_\ell, R_\ell), M_b + M_y\big).$$

Therefore,

$$\chi(\ell, b, y) = \mathrm{true} \iff A^{\min}_b \le A_\ell,\ E_\ell \le T_b,\ P_y \le P_b,\ D_y \le D_b.$$

Thus χ is the composite profunctor that maps each implementation triple to its feasibility truth value.

3. Composition of the subsystem profunctors

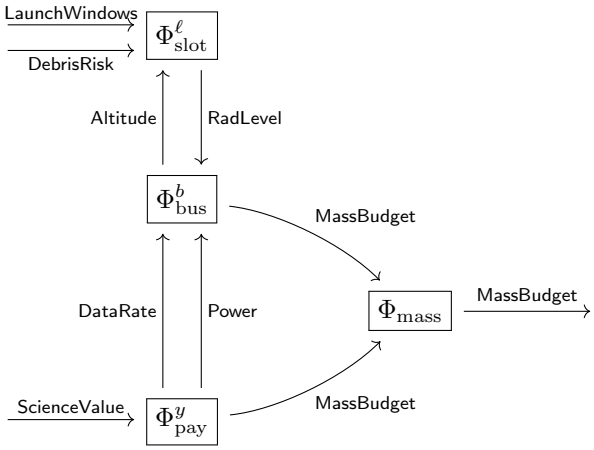
Arrows indicate the profunctor orientation from provided functionality to required resources. Fix an implementation triple $(\ell, b, y) \in D_{\mathrm{slot}} \times D_{\mathrm{bus}} \times D_{\mathrm{pay}}$. The corresponding co-design relation is obtained by first taking the parallel composition

$$\Phi^{\ell}_{\mathrm{slot}} \otimes \Phi^{b}_{\mathrm{bus}} \otimes \Phi^{y}_{\mathrm{pay}} \otimes \Phi_{\mathrm{mass}},$$

and then contracting the matching internal interfaces.

The Power, DataRate, and local MassBudget interfaces are connected by ordinary sequential composition. The coupled Altitude and RadLevel interfaces form a feedback loop, interpreted using the compact-closed

structure of Bool-enriched profunctors. The string diagram is presented below.



The arrows represent propagation from requested functionality to required resources in the profunctor orientation. The resulting external profunctor has type

$$\Phi^{\ell,b,y}_{\mathrm{total}} : \mathsf{ScienceValue} \times \mathsf{LaunchWindows} \times \mathsf{DebrisRisk} \nrightarrow \mathsf{MassBudget}.$$

Explicitly, it is given by the following long list of conditions.

$$\begin{aligned}
&\Phi^{\ell,b,y}_{\mathrm{total}}\big((s, W, \rho), m\big) = \mathrm{true} \\
&\qquad\Longleftrightarrow \\
&\qquad s \le S_y, \\
&\qquad W \subseteq W_\ell, \\
&\qquad \rho \le_{\mathrm{risk}} R_\ell, \\
&\qquad M_b + M_y \le m, \\
&\qquad A^{\min}_b \le A_\ell, \\
&\qquad E_\ell \le T_b, \\
&\qquad P_y \le P_b, \\
&\qquad D_y \le D_b.
\end{aligned}$$

Task 2

1. Feasible single-satellite configurations

Equip $D_{\mathrm{slot}},\ D_{\mathrm{bus}},\ D_{\mathrm{pay}}$ with the discrete preorder. For finite discrete preorders, a Bool-enriched profunctor is represented by a Boolean matrix.

By the formula for the composite resource-level profunctor obtained in Task 1, a catalogue triple (ℓ, b, y) is intrinsically feasible if and only if

$$A^{\min}_b \le A_\ell, \qquad E_\ell \le T_b, \qquad P_y \le P_b, \qquad D_y \le D_b.$$

The first two conditions concern slot–bus compatibility, while the last two concern bus–payload compatibility. We therefore factor the feasibility predicate into the two Boolean relations defined below.

Slot–bus compatibility matrix

Define the Bool-profunctor $K : D_{\text{slot}} \nrightarrow D_{\text{bus}}$ by

$$K_{\ell b} = 1 \quad \Longleftrightarrow \quad A_b^{\min} \leq A_\ell \text{ and } E_\ell \leq T_b.$$

Using the row order $L1, L2, L3, L4$ and the column order μSat, SmallSat-A, SmallSat-B, its Boolean matrix is

$$K = \begin{array}{c|ccc} & \mu\text{Sat} & \text{SmallSat-A} & \text{SmallSat-B} \\ \hline L1 & 0 & 1 & 1 \\ L2 & 0 & 0 & 1 \\ L3 & 1 & 1 & 0 \\ L4 & 0 & 1 & 1 \end{array}.$$

For example, the entry $K_{L2,\text{SmallSat-A}} = 0$ because the altitude condition holds, but the radiation condition does not, i.e. $400 \leq 550$, but High $\not\leq$ Medium.

Bus–payload compatibility matrix

Define the Bool-profunctor $L : D_{\text{bus}} \nrightarrow D_{\text{pay}}$ by

$$L_{by} = 1 \quad \Longleftrightarrow \quad P_y \leq P_b \text{ and } D_y \leq D_b.$$

Using the payload order Multispectral, Hyperspectral, SAR, Thermal, the matrix is

$$L = \begin{array}{c|cccc} & \text{Multispectral} & \text{Hyperspectral} & \text{SAR} & \text{Thermal} \\ \hline \mu\text{Sat} & 0 & 0 & 0 & 0 \\ \text{SmallSat-A} & 1 & 0 & 0 & 1 \\ \text{SmallSat-B} & 1 & 1 & 1 & 1 \end{array}$$

Although μSat provides sufficient power for the Multispectral and Thermal payloads, its downlink capacity is only 10 Mbps, whereas these payloads require 20 Mbps and 15 Mbps. Hence the entire μSat row is zero.

Existential composition

The ordinary Boolean product $K \odot L : D_{\text{slot}} \nrightarrow D_{\text{pay}}$ has entries

$$(K \odot L)_{\ell y} = \bigvee_{b \in D_{\text{bus}}} (K_{\ell b} \wedge L_{by}).$$

Thus $(K \odot L)_{\ell y} = 1$ means that there exists at least one bus which makes slot ℓ compatible with payload y. Explicitly,

$$K \odot L = \begin{array}{c|cccc} & \text{Multispectral} & \text{Hyperspectral} & \text{SAR} & \text{Thermal} \\ \hline L1 & 1 & 1 & 1 & 1 \\ L2 & 1 & 1 & 1 & 1 \\ L3 & 1 & 0 & 0 & 1 \\ L4 & 1 & 1 & 1 & 1 \end{array}.$$

This product eliminates the intermediate bus index. It therefore determines whether some suitable bus exists, but it does not by itself list the feasible triples (ℓ, b, y).

Retaining the bus choice

To record the bus choice, define the feasibility relation $\chi : D_{\text{slot}} \times D_{\text{bus}} \times D_{\text{pay}} \nrightarrow \mathbf{1}$ by

$$\chi(\ell, b, y) = K_{\ell b} \wedge L_{by}.$$

Equivalently, after grouping the first two indices, define the Boolean matrix

$$F : D_{\text{slot}} \times D_{\text{bus}} \nrightarrow D_{\text{pay}}, \qquad F_{(\ell,b),y} = K_{\ell b} \wedge L_{by}.$$

Thus $F_{(\ell,b),y} = 1$ precisely when (ℓ, b, y) is a feasible single-satellite configuration.

The feasibility matrix is

$F =$

(ℓ, b)	Multispectral	Hyperspectral	SAR	Thermal
$(L1, \mu\text{Sat})$	0	0	0	0
$(L1, \text{SmallSat-A})$	1	0	0	1
$(L1, \text{SmallSat-B})$	1	1	1	1
$(L2, \mu\text{Sat})$	0	0	0	0
$(L2, \text{SmallSat-A})$	0	0	0	0
$(L2, \text{SmallSat-B})$	1	1	1	1
$(L3, \mu\text{Sat})$	0	0	0	0
$(L3, \text{SmallSat-A})$	1	0	0	1
$(L3, \text{SmallSat-B})$	0	0	0	0
$(L4, \mu\text{Sat})$	0	0	0	0
$(L4, \text{SmallSat-A})$	1	0	0	1
$(L4, \text{SmallSat-B})$	1	1	1	1

.

The number of feasible configurations is the number of entries equal to 1:

$$2 + 4 + 4 + 2 + 2 + 4 = 18.$$

The feasible set is therefore

$$\begin{aligned}\mathcal{F} = {} & \{L1, L3, L4\} \times \{\text{SmallSat-A}\} \times \{\text{Multispectral}, \text{Thermal}\} \\ & \cup \{L1, L2, L4\} \times \{\text{SmallSat-B}\} \times \{\text{Multispectral}, \text{Hyperspectral}, \text{SAR}, \text{Thermal}\}.\end{aligned}$$

Under the convention introduced in Task 1, the composite is evaluated at the implementation's own total mass requirement

$$m = M_b + M_y.$$

The corresponding mass condition is therefore

$$M_b + M_y \le m,$$

which holds with equality. It follows that mass is recorded as an external resource requirement but does not alter the intrinsic Boolean feasibility predicate χ. If instead an external mass limit $m_{\max}$ were imposed, the matrix would be filtered by the additional condition

$$M_b + M_y \le m_{\max}.$$

Feasible example

Consider $(L1, \text{SmallSat-A}, \text{Multispectral})$. The relevant matrix entry is

$$\begin{aligned}\chi(L1, \text{SmallSat-A}, \text{Multispectral}) &= K_{L1,\text{SmallSat-A}} \wedge L_{\text{SmallSat-A,Multispectral}} \\ &= 1 \wedge 1 \\ &= 1.\end{aligned}$$

We can check that

$$400 \le 550, \qquad \text{Medium} \le \text{Medium}, \qquad 35 \le 120, \qquad 20 \le 50.$$

Thus the configuration is feasible. Its total mass requirement is $80 + 8 = 88$ kg.

Infeasible example: radiation mismatch

Consider $(L2, \text{SmallSat-A}, \text{Multispectral})$. Here $K_{L2,\text{SmallSat-A}} = 0$ because High $\not\leq$ Medium. Therefore

$$\begin{aligned}\chi(L2, \text{SmallSat-A}, \text{Multispectral}) &= K_{L2,\text{SmallSat-A}} \wedge L_{\text{SmallSat-A,Multispectral}} \\ &= 0 \wedge 1 \\ &= 0.\end{aligned}$$

Infeasible example: insufficient downlink

Consider $(L1, \text{SmallSat-A}, \text{Hyperspectral})$. The slot and bus are compatible, so $K_{L1,\text{SmallSat-A}} = 1$. However, $L_{\text{SmallSat-A,Hyperspectral}} = 0$ because the Hyperspectral payload requires 80 Mbps while SmallSat-A provides only 50 Mbps. Thus

$$\begin{aligned}\chi(L1, \text{SmallSat-A}, \text{Hyperspectral}) &= 1 \wedge 0 \\ &= 0.\end{aligned}$$

2. Convenient Three-Satellite Constellations

Let $\mathcal{U} = \{U_{123}, U_{124}, U_{134}, U_{234}\}$ be the set of three-element subsets of D_{slot}, where

$$\begin{aligned}U_{123} &= \{L1, L2, L3\}, \quad U_{124} = \{L1, L2, L4\}, \\ U_{134} &= \{L1, L3, L4\}, \quad U_{234} = \{L2, L3, L4\}.\end{aligned}$$

Because each slot is used at most once, every convenient constellation uses one of these four slot sets.

Launch-window relation

Equip $Q = \{Q_1, Q_2, Q_3\}$ and $\mathcal{U} = \{U_{123}, U_{124}, U_{134}, U_{234}\}$ with the discrete preorder. Define the Boolean slot–window relation

$$W : D_{\text{slot}} \nrightarrow Q, \qquad W_{\ell q} = 1 \iff q \in W_\ell.$$

Using the row order $L1, L2, L3, L4$ and the column order Q_1, Q_2, Q_3, the matrix of W is

$$W = \begin{array}{c|ccc} & Q_1 & Q_2 & Q_3 \\ \hline L1 & 1 & 1 & 0 \\ L2 & 1 & 0 & 0 \\ L3 & 1 & 1 & 1 \\ L4 & 0 & 1 & 1 \end{array}.$$

For $U \in \mathcal{U}$, define

$$J : \mathcal{U} \nrightarrow Q, \qquad J_{Uq} = \bigwedge_{\ell \in U} W_{\ell q}.$$

Thus $J_{Uq} = 1$ precisely when q is a launch window common to every slot in U. Explicitly,

$$J = \begin{array}{c|ccc} & Q_1 & Q_2 & Q_3 \\ \hline U_{123} & 1 & 0 & 0 \\ U_{124} & 0 & 0 & 0 \\ U_{134} & 0 & 1 & 0 \\ U_{234} & 0 & 0 & 0 \end{array}.$$

Let $\top_Q : Q \nrightarrow \mathbf{1}$, $(\top_Q)_{q,*} = 1$, be the all-true relation. In matrix form,

$$\top_Q = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

The common-launch-window predicate is the composite

$$\mathbf{c}_{\text{window}} = J \odot \top_Q : \mathcal{U} \nrightarrow \mathbf{1}.$$

Its matrix is

$$\mathbf{c}_{\text{window}} = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix},$$

because

$$(\mathbf{c}_{\text{window}})_{U,*} = \bigvee_{q \in Q} J_{Uq} = \bigvee_{q \in Q} \bigwedge_{\ell \in U} W_{\ell q}.$$

Hence $(\mathbf{c}_{\text{window}})_{U,*} = 1$ if and only if the slots in U share at least one common launch window.

Debris-risk constraint

The total debris risks of the four slot sets are

Slot set	Total debris risk
U_{123}	$0.003 + 0.001 + 0.008 = 0.012$
U_{124}	$0.003 + 0.001 + 0.002 = 0.006$
U_{134}	$0.003 + 0.008 + 0.002 = 0.013$
U_{234}	$0.001 + 0.008 + 0.002 = 0.011$

.

Define the Boolean debris-risk predicate $\mathbf{c}_{\text{risk}} : \mathcal{U} \nrightarrow \mathbf{1}$ by

$$(\mathbf{c}_{\text{risk}})_{U,*} = 1 \quad \Longleftrightarrow \quad \sum_{\ell \in U} R_\ell \leq 0.012.$$

Its matrix is

$$\mathbf{c}_{\text{risk}} = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 1 \end{pmatrix}.$$

The combined slot predicate is the pointwise conjunction

$$\mathbf{c}_{\text{slot}} = \mathbf{c}_{\text{window}} \wedge \mathbf{c}_{\text{risk}} : \mathcal{U} \nrightarrow \mathbf{1}.$$

Thus

$$\mathbf{c}_{\text{slot}} = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} \wedge \begin{pmatrix} 1 \\ 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

Hence the unique slot set satisfying both the common-launch-window and debris-risk constraints is

$$U_{123} = \{L1, L2, L3\}.$$

These slots share the launch window Q_1, and their total debris risk is exactly 0.012.

Parallel composition of the single-satellite matrices

Write

$$\mu = \mu\text{Sat}, \qquad A = \text{SmallSat-A}, \qquad B = \text{SmallSat-B},$$

and abbreviate the payloads by

$$M = \text{Multispectral}, \qquad H = \text{Hyperspectral}, \qquad S = \text{SAR}, \qquad T = \text{Thermal}.$$

Their science values are

$$v(M) = 3, \qquad v(H) = 7, \qquad v(S) = 10, \qquad v(T) = 4.$$

From the single-satellite feasibility matrix F, extract the slot-specific matrices

$$F_\ell(b, y) = \chi(\ell, b, y).$$

For the three selected slots, these are

$$F_{L1} = \begin{array}{c|cccc} & M & H & S & T \\ \hline \mu & 0 & 0 & 0 & 0 \\ A & 1 & 0 & 0 & 1 \\ B & 1 & 1 & 1 & 1 \end{array},$$

$$F_{L2} = \begin{array}{c|cccc} & M & H & S & T \\ \hline \mu & 0 & 0 & 0 & 0 \\ A & 0 & 0 & 0 & 0 \\ B & 1 & 1 & 1 & 1 \end{array},$$

$$F_{L3} = \begin{array}{c|cccc} & M & H & S & T \\ \hline \mu & 0 & 0 & 0 & 0 \\ A & 1 & 0 & 0 & 1 \\ B & 0 & 0 & 0 & 0 \end{array}.$$

By Theorem 4.63 in [1], parallel composition is the product of Bool-categories with pointwise tensor on profunctors. For finite discrete implementation categories, its Boolean-matrix representation is the Kronecker product. Hence the parallel composite is represented by

$$G = F_{L1} \otimes F_{L2} \otimes F_{L3}.$$

Its rows are indexed by bus triples (b_1, b_2, b_3), and its columns are indexed by payload triples (y_1, y_2, y_3). Its entries are

$$G_{(b_1,b_2,b_3),(y_1,y_2,y_3)} = F_{L1}(b_1, y_1) \wedge F_{L2}(b_2, y_2) \wedge F_{L3}(b_3, y_3).$$

Thus an entry of G is 1 precisely when the three selected single-satellite configurations are all feasible.

Before imposing the science constraint, G has

$$6 \cdot 4 \cdot 2 = 48$$

entries equal to 1, since $L1$, $L2$, and $L3$ admit, respectively, 6, 4, and 2 feasible bus–payload choices.

Science-threshold predicate

Define a Boolean matrix V, with the same row and column indexing as G, by

$$V_{(b_1,b_2,b_3),(y_1,y_2,y_3)} = 1 \quad \Longleftrightarrow \quad v(y_1) + v(y_2) + v(y_3) \geq 18.$$

The matrix of convenient constellations is the pointwise conjunction $C = G \wedge V$.

Hence $C_{(b_1,b_2,b_3),(y_1,y_2,y_3)} = 1$ if and only if the three corresponding configurations are feasible and their total science value is at least 18.

Although G and C are indexed using the ordered slot list $(L1, L2, L3)$, each nonzero entry corresponds to exactly one multiset of configurations. Indeed, the three slot labels are distinct, so any such multiset has a unique ordering by its slot coordinate.

There is no restriction on reusing a bus design or payload type; only orbital slots must be distinct.

Nonzero entries of the constellation matrix

Since $L3$ supports only the payloads M and T, the support of C can be computed in two cases.

Case 1: $L3$ carries M. The contribution from $L3$ is 3, so the payloads on $L1$ and $L2$ must contribute at least

$$18 - 3 = 15.$$

The only payload triples occurring in the support of C are

$$(H, S, M), \qquad (S, H, M), \qquad (S, S, M).$$

The corresponding bus triple is (B, B, A) in each case. Therefore the three constellations are

$$\begin{aligned}
\mathcal{C}_1 &= \{\{(L1, B, H), (L2, B, S), (L3, A, M)\}\},\\
\mathcal{C}_2 &= \{\{(L1, B, S), (L2, B, H), (L3, A, M)\}\},\\
\mathcal{C}_3 &= \{\{(L1, B, S), (L2, B, S), (L3, A, M)\}\}.
\end{aligned}$$

Their science values are

$$20, \qquad 20, \qquad 23.$$

Case 2: $L3$ carries T. The contribution from $L3$ is 4, so the payloads on $L1$ and $L2$ must contribute at least

$$18 - 4 = 14.$$

The payload triples occurring in the support of C are

$$(T, S, T), \qquad (H, H, T), \qquad (H, S, T), \qquad (S, T, T), \qquad (S, H, T), \qquad (S, S, T).$$

For the triple (T, S, T), the Thermal payload on $L1$ may use either bus A or bus B. All remaining choices require bus B on $L1$. The bus on $L2$ must be B, and the bus on $L3$ must be A.

This gives seven nonzero entries:

$$\begin{aligned}
\mathcal{C}_4 &= \{\{(L1, A, T), (L2, B, S), (L3, A, T)\}\},\\
\mathcal{C}_5 &= \{\{(L1, B, T), (L2, B, S), (L3, A, T)\}\},\\
\mathcal{C}_6 &= \{\{(L1, B, H), (L2, B, H), (L3, A, T)\}\},\\
\mathcal{C}_7 &= \{\{(L1, B, H), (L2, B, S), (L3, A, T)\}\},\\
\mathcal{C}_8 &= \{\{(L1, B, S), (L2, B, T), (L3, A, T)\}\},\\
\mathcal{C}_9 &= \{\{(L1, B, S), (L2, B, H), (L3, A, T)\}\},\\
\mathcal{C}_{10} &= \{\{(L1, B, S), (L2, B, S), (L3, A, T)\}\}.
\end{aligned}$$

Their total science values are

$$18, \qquad 18, \qquad 18, \qquad 21, \qquad 18, \qquad 21, \qquad 24.$$

The matrix C therefore has $3 + 7 = 10$ entries equal to 1. Hence the number of convenient three-satellite constellations is 10.

3. Pareto-optimal convenient constellations

For a convenient constellation $\mathcal{C}$, let $V(\mathcal{C})$ denote its total science value and let $R(\mathcal{C})$ denote its total debris risk.

A convenient constellation $\mathcal{C}'$ dominates $\mathcal{C}$ if

$$V(\mathcal{C}') \geq V(\mathcal{C}), \qquad R(\mathcal{C}') \leq R(\mathcal{C}),$$

with at least one inequality strict.

From Task 2.2, every convenient constellation must use exactly the three slots $\{L1, L2, L3\}$. It follows that every convenient constellation has the same total debris risk:

$$R(\mathcal{C}) = 0.003 + 0.001 + 0.008 = 0.012$$

Therefore, among convenient constellations, Pareto dominance is determined entirely by total science value: a constellation with larger science value dominates any constellation with smaller science value, since their debris risks are equal.

The science values of the ten convenient constellations are

$\mathcal{C}$	$\mathcal{C}_1$	$\mathcal{C}_2$	$\mathcal{C}_3$	$\mathcal{C}_4$	$\mathcal{C}_5$	$\mathcal{C}_6$	$\mathcal{C}_7$	$\mathcal{C}_8$	$\mathcal{C}_9$	$\mathcal{C}_{10}$
$V(\mathcal{C})$	20	20	23	18	18	18	21	18	21	24

The unique maximum science value is 24, attained by

$$\mathcal{C}_{10} = \{\{(L1, \text{SmallSat-B}, \text{SAR}), (L2, \text{SmallSat-B}, \text{SAR}), (L3, \text{SmallSat-A}, \text{Thermal})\}\}.$$

Every other convenient constellation has the same debris risk but a strictly smaller science value, and is therefore dominated by $\mathcal{C}_{10}$. Hence $\mathcal{C}_{10}$ is the unique Pareto-optimal convenient constellation.

Task 3: Upgrades

1. Effect of the upgrades on feasible configurations

Recall the Boolean compatibility matrices $K : D_{\text{slot}} \nrightarrow D_{\text{bus}}$, $L : D_{\text{bus}} \nrightarrow D_{\text{pay}}$, where

$$K_{\ell b} = 1 \iff A_b^{\min} \le A_\ell \text{ and } E_\ell \le T_b,$$

$$L_{by} = 1 \iff P_y \le P_b \text{ and } D_y \le D_b.$$

Using

$$\mu = \mu\text{Sat}, \qquad A = \text{SmallSat-A}, \qquad B = \text{SmallSat-B},$$

and

$$M = \text{Multispectral}, \quad H = \text{Hyperspectral}, \quad S = \text{SAR}, \quad T = \text{Thermal},$$

the original matrices are

$$K = \begin{array}{c|ccc} & \mu & A & B \\ \hline L1 & 0 & 1 & 1 \\ L2 & 0 & 0 & 1 \\ L3 & 1 & 1 & 0 \\ L4 & 0 & 1 & 1 \end{array} \qquad L = \begin{array}{c|cccc} & M & H & S & T \\ \hline \mu & 0 & 0 & 0 & 0 \\ A & 1 & 0 & 0 & 1 \\ B & 1 & 1 & 1 & 1 \end{array}.$$

The feasibility relation on triples is

$$F((\ell, b), y) = K_{\ell b} \wedge L_{by}.$$

Before either upgrade, the number of nonzero entries of F is $|\mathcal{F}| = 18$.

Upgrade X

Under Upgrade X, SmallSat-A's radiation tolerance changes from Medium to High. Therefore the slot–bus compatibility matrix becomes

$$K_X = \begin{array}{c|ccc} & \mu & A & B \\ \hline L1 & 0 & 1 & 1 \\ L2 & 0 & 1 & 1 \\ L3 & 1 & 1 & 0 \\ L4 & 0 & 1 & 1 \end{array}.$$

The only changed entry is

$$(K_X)_{L2,A} = 1,$$

because

$$A_A^{\min} = 400 \le 550 = A_{L2}, \qquad E_{L2} = \text{High} \le \text{High} = T_A^X.$$

The bus–payload matrix L is unchanged. Hence

$$F_X((\ell, b), y) = (K_X)_{\ell b} \wedge L_{by}.$$

The previously zero row corresponding to $(L2, A)$ becomes

$$(0, 0, 0, 0) \longmapsto (1, 0, 0, 1).$$

Thus Upgrade X adds exactly the configurations $(L2, A, M)$ and $(L2, A, T)$.
Therefore $\mathcal{F}_X = \mathcal{F} \cup \{(L2, A, M), (L2, A, T)\}$, and $|\mathcal{F}_X| = 20$.

Upgrade Y

Under Upgrade Y, the power capacity of μSat changes from 40 W to 70 W.
Its power compatibility with the payloads becomes

	M	H	S	T
$P_y \leq 70$	1	0	0	1

.

However, its downlink capacity remains 10 Mbps:

	M	H	S	T
$D_y \leq 10$	0	0	0	0.

Since bus–payload feasibility is the conjunction of the power and downlink conditions, the μSat row remains

$$(1, 0, 0, 1) \wedge (0, 0, 0, 0) = (0, 0, 0, 0).$$

Consequently, $L_Y = L,\ F_Y = F$, and therefore $\mathcal{F}_Y = \mathcal{F},\ |\mathcal{F}_Y| = 18$.
Hence Upgrade X adds more feasible single-satellite configurations.

2. Upgrades as morphisms

Define $\mathcal{B} = \mathsf{Power} \times \mathsf{DataRate} \times \mathsf{RadLevel} \times \mathsf{Altitude}^{\mathrm{op}} \times \mathsf{MassBudget}^{\mathrm{op}}$.
An object of $\mathcal{B}$ is a bus profile

$$\beta = (P, D, T, A^{\min}, M).$$

The first three coordinates record resources provided by the bus, whereas $A^{\min}$ and M record required resources. The opposite orders on $\mathsf{Altitude}$ and $\mathsf{MassBudget}$ express that a smaller minimum-altitude requirement and a smaller mass requirement constitute improvements.
There is a morphism

$$\beta \longrightarrow \beta' \iff P \leq P',\ D \leq D',\ T \leq T',\ A'^{\min} \leq A^{\min},\ M' \leq M.$$

Thus a morphism represents a componentwise weak improvement: the upgraded bus provides at least as much power, downlink capacity, and radiation tolerance, while requiring no greater minimum altitude or mass budget. Since this relation is reflexive and transitive, $\mathcal{B}$ is a preorder and hence a thin category.
For Upgrade X, define

$$\beta_A^0 = (120, 50, \mathrm{Medium}, 400, 80)$$
$$\beta_A^X = (120, 50, \mathrm{High}, 400, 80).$$

Since Medium $\leq$ High and all other coordinates are unchanged, there is a morphism

$$u_X : \beta_A^0 \longrightarrow \beta_A^X.$$

For Upgrade Y, define

$$\beta_\mu^0 = (40, 10, \mathrm{Low}, 350, 25)$$
$$\beta_\mu^Y = (70, 10, \mathrm{Low}, 350, 25).$$

Since $40 \leq 70$ and all other coordinates are unchanged, there is a morphism

$$u_Y : \beta_\mu^0 \longrightarrow \beta_\mu^Y.$$

More generally, let

$$\mathcal{C} = \mathcal{B}^{D_{\mathrm{bus}}}$$

be the functor category from the discrete category D_{bus} to $\mathcal{B}$. An object

$$C : D_{\mathrm{bus}} \longrightarrow \mathcal{B}$$

assigns a bus profile to each fixed bus label and therefore represents a bus catalogue.

Since D_{bus} is discrete, a natural transformation

$$U : C \longrightarrow C'$$

exists precisely when there is a morphism

$$C(b) \longrightarrow C'(b)$$

in $\mathcal{B}$ for every $b \in D_{\mathrm{bus}}$. Thus morphisms in $\mathcal{C}$ are componentwise monotone upgrades of a fixed bus catalogue.

Let C_0 denote the original catalogue:

$$\begin{aligned} C_0(\mu) &= (40, 10, \mathrm{Low}, 350, 25), \\ C_0(A) &= (120, 50, \mathrm{Medium}, 400, 80), \\ C_0(B) &= (200, 150, \mathrm{High}, 500, 150). \end{aligned}$$

Define C_X to agree with C_0 except that

$$C_X(A) = \beta_A^X,$$

and define C_Y to agree with C_0 except that

$$C_Y(\mu) = \beta_\mu^Y.$$

The upgrades are natural transformations

$$U_X : C_0 \longrightarrow C_X, \qquad U_Y : C_0 \longrightarrow C_Y,$$

whose components are

$$(U_X)_A = u_X, \qquad (U_X)_\mu = \mathrm{id}, \qquad (U_X)_B = \mathrm{id},$$

and

$$(U_Y)_\mu = u_Y, \qquad (U_Y)_A = \mathrm{id}, \qquad (U_Y)_B = \mathrm{id},$$

respectively.

Define the preorder of Boolean feasibility matrices by

$$\mathcal{M} = \mathbf{Bool}^{(D_{\mathrm{slot}} \times D_{\mathrm{bus}})^{\mathrm{op}} \times D_{\mathrm{pay}}},$$

ordered pointwise. Since the implementation preorders are discrete, an object of $\mathcal{M}$ may be represented as a Boolean matrix indexed by slot–bus pairs and payloads.

There is a feasibility-semantics functor

$$\mathcal{S} : \mathcal{C} \longrightarrow \mathcal{M}.$$

For a catalogue C, define

$$\mathcal{S}(C)_{(\ell,b),y} = 1 \iff A_{C(b)}^{\min} \leq A_\ell,\ E_\ell \leq T_{C(b)},\ P_y \leq P_{C(b)},\ D_y \leq D_{C(b)}.$$

This assignment is monotone. Suppose $C \longrightarrow C'$ is a componentwise upgrade and that $\mathcal{S}(C)_{(\ell,b),y} = 1$.

Since $C(b) \to C'(b)$ in $\mathcal{B}$, we have

$$A^{\min}_{C'(b)} \le A^{\min}_{C(b)} \le A_\ell,$$
$$E_\ell \le T_{C(b)} \le T_{C'(b)},$$
$$P_y \le P_{C(b)} \le P_{C'(b)},$$
$$D_y \le D_{C(b)} \le D_{C'(b)}.$$

Therefore $\mathcal{S}(C')_{(\ell,b),y} = 1$. It follows that $\mathcal{S}(C) \le \mathcal{S}(C')$ pointwise, so $\mathcal{S}$ is a monotone map and therefore a functor between the corresponding thin categories.

For Upgrade X,

$$F = \mathcal{S}(C_0) < \mathcal{S}(C_X) = F_X,$$

and F_X contains exactly two additional nonzero entries.

For Upgrade Y,

$$F = \mathcal{S}(C_0) = \mathcal{S}(C_Y) = F_Y.$$

Thus U_Y is a nonidentity morphism in the design category, even though its image under the present feasibility semantics is unchanged.

Task 4: Time-varying feasibility

1. Categorical structure

We model a finite time horizon as the thin category

$$\mathbf{T}_N = ([N], \le),$$

whose objects are $0, 1, \ldots, N$ and which has a unique morphism

$$t \longrightarrow u \quad \Longleftrightarrow \quad t \le u.$$

Ambient state spaces

Let

$$\Sigma = \{\perp\} \sqcup \big(D_{\text{bus}} \times D_{\text{pay}} \times \mathbb{N}\big)$$

be the set of local slot states, equipped with the discrete preorder, and let

$$\overline{X} = \Sigma^{D_{\text{slot}}}.$$

For every $t \in \mathbf{T}_N$, set

$$X_t = \overline{X}.$$

Thus a state $x_t \in X_t$ is a function

$$x_t : D_{\text{slot}} \longrightarrow \Sigma.$$

For a slot $\ell \in D_{\text{slot}}$, the value $x_t(\ell) = \perp$ means that the slot is empty, while

$$x_t(\ell) = (b, y, a)$$

means that slot ℓ contains an active satellite with bus b, payload y, and age a. This representation automatically ensures that each slot is occupied by at most one active satellite.

Once an initial state x_0, or more generally a set of admissible initial states, is specified, the transition relations introduced below determine which ambient states are reachable at each later time.

Time-dependent subsystem data

We interpret "drops by 10% each year" as retaining 90% of the previous year's power. Thus a bus b of age a has effective payload-support power

$$P_b^{\mathrm{eff}}(a) = 0.9^a P_b.$$

The debris risk of slot ℓ at absolute time t is

$$R_\ell(t) = R_\ell(0) + 0.001t.$$

Hence replacing a satellite resets its age, and therefore its power degradation, but does not reset the debris-risk growth associated with the orbital slot.

For an active satellite occupying slot ℓ, define the local compatibility predicate

$$\chi^{\mathrm{sat}}(\ell, b, y, a) = [A_b^{\min} \le A_\ell] \wedge [E_\ell \le T_b] \wedge [P_y \le 0.9^a P_b] \wedge [D_y \le D_b],$$

where $[Q] \in \mathbf{Bool}$ denotes the truth value of the proposition Q.

For a state $x \in X_t$, define the set of occupied slots by

$$\mathrm{Act}(x) = \{\ell \in D_{\mathrm{slot}} \mid x(\ell) \neq \bot\}.$$

If $x(\ell) = (b_\ell, y_\ell, a_\ell)$ for every $\ell \in \mathrm{Act}(x)$, define the time-t feasibility relation

$$\Phi_t : X_t \nrightarrow \mathbf{1}$$

by

$$\begin{aligned}\Phi_t(x) = \big[|\,\mathrm{Act}(x)| = 3\big] &\wedge \left[\bigcap_{\ell \in \mathrm{Act}(x)} W_\ell \neq \varnothing\right] \\ &\wedge \bigwedge_{\ell \in \mathrm{Act}(x)} \chi^{\mathrm{sat}}(\ell, b_\ell, y_\ell, a_\ell) \\ &\wedge \left[\sum_{\ell \in \mathrm{Act}(x)} S_{y_\ell} \ge 18\right] \\ &\wedge \left[\sum_{\ell \in \mathrm{Act}(x)} R_\ell(t) \le 0.012\right].\end{aligned}$$

Because X_t is discrete, the Bool-enriched profunctor $\Phi_t : X_t \nrightarrow \mathbf{1}$ is simply a Boolean predicate on states. The equality $\Phi_t(x) = 1$ means that x is a convenient three-satellite constellation at time t.

We retain the common-launch-window condition as a static compatibility criterion for every active three-satellite constellation. Operational restrictions on whether a launch or replacement may occur at a particular time are encoded separately in the transition relation below.

Reconfiguration actions

Define the set of local candidate actions by

$$\begin{aligned}\mathcal{U}_{\mathrm{loc}} = &\{\mathrm{idle}, \mathrm{keep}, \mathrm{decommission}\} \\ &\sqcup \{\mathrm{launch}(b, y) \mid (b, y) \in D_{\mathrm{bus}} \times D_{\mathrm{pay}}\} \\ &\sqcup \{\mathrm{replace}(b, y) \mid (b, y) \in D_{\mathrm{bus}} \times D_{\mathrm{pay}}\}.\end{aligned}$$

For every $t < N$, let

$$U_t \subseteq \mathcal{U}_{\mathrm{loc}}^{D_{\mathrm{slot}}}$$

be the discrete preorder of candidate global action profiles at time t.

We interpret u_t as an action performed in the transition from time t to time $t+1$. A retained satellite is one year older in x_{t+1}, whereas a newly launched or replacement satellite enters x_{t+1} with age 0. Feasibility is evaluated only at the discrete observation times $0, 1, \ldots, N$, so a replacement is treated as an atomic transition.

Define the local transition relation

$$\tau : \Sigma \times \mathcal{U}_{\mathrm{loc}} \nrightarrow \Sigma$$

by declaring $\tau(s, u, s') = 1$ precisely in the following cases:

s	u	s'
$\bot$	idle	$\bot$
(b, y, a)	keep	$(b, y, a+1)$
(b, y, a)	decommission	$\bot$
$\bot$	launch(b', y')	$(b', y', 0)$
(b, y, a)	replace(b', y')	$(b', y', 0)$.

All other local state–action–successor triples have truth value 0.

Let

$$G_t : X_t \times U_t \times X_{t+1} \longrightarrow \mathbf{Bool}$$

be a global admissibility predicate. It may encode, for example, launch capacity, operational launch-window availability, cost budgets, or a bound on the number of replacements performed during one time step.

The controlled transition profunctor

$$\Gamma_t : X_t \times U_t \nrightarrow X_{t+1}$$

is then defined by

$$\Gamma_t((x, u), x') = \left(\bigwedge_{\ell \in D_{\mathrm{slot}}} \tau\big(x(\ell), u(\ell), x'(\ell)\big) \right) \wedge G_t(x, u, x').$$

Thus $\Gamma_t((x, u), x') = 1$ precisely when every local transition is valid and the global action constraints are satisfied.

If the particular action is not externally relevant, we existentially quantify it out and define

$$A_t : X_t \nrightarrow X_{t+1}$$

by

$$A_t(x, x') = \bigvee_{u \in U_t} \Gamma_t((x, u), x').$$

Thus $A_t(x, x') = 1$ if and only if there exists an admissible reconfiguration action profile taking x at time t to x' at time $t+1$.

Ambient and feasible transition diagrams

Since every X_t is discrete, the state spaces and transition profunctors may be regarded as objects and morphisms in the category **Rel** of sets and relations. Equivalently, they lie in the full subcategory of the category of preorders and Bool-enriched profunctors spanned by discrete preorders.

For relations $R : X \nrightarrow Y$ and $S : Y \nrightarrow Z$, write

$$(R \odot S)(x, z) = \bigvee_{y \in Y} \big(R(x, y) \wedge S(y, z)\big)$$

for relational composition in the left-to-right convention used below.

For every t, define

$$A_{t,t} = \Delta_{X_t},$$

where Δ_{X_t} is the identity relation on X_t. For $t < u$, define

$$A_{t,u} = A_t \odot A_{t+1} \odot \cdots \odot A_{u-1} : X_t \nrightarrow X_u.$$

Explicitly,

$$A_{t,u}(x_t, x_u) = \bigvee_{x_{t+1},\ldots,x_{u-1}} \bigwedge_{k=t}^{u-1} A_k(x_k, x_{k+1}).$$

Hence $A_{t,u}(x_t, x_u) = 1$ if and only if there exists a sequence of intermediate states and admissible actions taking x_t to x_u.

The assignments

$$t \longmapsto X_t, \qquad (t \le u) \longmapsto A_{t,u}$$

define an ambient transition functor

$$\mathcal{X} : \mathbf{T}_N \longrightarrow \mathbf{Rel},$$

because

$$A_{t,t} = \Delta_{X_t}, \qquad A_{t,u} \odot A_{u,v} = A_{t,v} \quad (t \le u \le v).$$

The time-indexed feasibility problem is the package consisting of this ambient transition functor, the controlled relations Γ_t, and the family of feasibility predicates

$$\Phi_t : X_t \nrightarrow \mathbf{1}.$$

The functor $\mathcal{X}$ alone describes physically admissible ambient transitions; the predicates Φ_t specify which states are feasible at each time.

For the purpose of studying trajectories that remain feasible, feasibility may instead be internalized into the state spaces. Define

$$X_t^{\mathrm{feas}} = \{x \in X_t \mid \Phi_t(x) = 1\}$$

and restrict A_t to

$$A_t^{\mathrm{feas}} : X_t^{\mathrm{feas}} \nrightarrow X_{t+1}^{\mathrm{feas}}.$$

The composites of the relations A_t^{feas} define a functor

$$\mathcal{X}^{\mathrm{feas}} : \mathbf{T}_N \longrightarrow \mathbf{Rel}$$

whose arrows represent precisely the state evolutions that remain feasible at every intermediate time.

Multi-period feasibility

A trajectory over the horizon $0, \ldots, N$ consists of states and actions

$$x_0, u_0, x_1, u_1, \ldots, u_{N-1}, x_N.$$

It is feasible if

$$\Phi_t(x_t) = 1 \qquad \text{for every } 0 \le t \le N,$$

and

$$\Gamma_t((x_t, u_t), x_{t+1}) = 1 \qquad \text{for every } 0 \le t < N.$$

For a fixed initial state $x_0 \in X_0$, the existence of a feasible trajectory is described by the Boolean predicate

$$\mathrm{Feas}_N(x_0) = \bigvee_{\substack{u_0,\ldots,u_{N-1} \\ x_1,\ldots,x_N}} \left(\bigwedge_{t=0}^{N} \Phi_t(x_t) \right) \wedge \left(\bigwedge_{t=0}^{N-1} \Gamma_t((x_t, u_t), x_{t+1}) \right).$$

This predicate is true exactly when there exists a sequence of reconfiguration actions that keeps the constellation feasible throughout the entire horizon.

Backward viability recursion

It is useful to distinguish instantaneous feasibility from future viability. Define predicates $V_t : X_t \nrightarrow \mathbf{1}$ backwards in time. At the terminal time, let $V_N = \Phi_N$. For $t < N$, define

$$V_t(x) = \Phi_t(x) \wedge \bigvee_{\substack{u \in U_t \\ x' \in X_{t+1}}} \left(\Gamma_t((x,u),x') \wedge V_{t+1}(x')\right).$$

After existentially eliminating the action variable, this becomes

$$V_t = \Phi_t \wedge (A_t \odot V_{t+1}).$$

This expression is well typed because

$$A_t : X_t \nrightarrow X_{t+1}, \qquad V_{t+1} : X_{t+1} \nrightarrow \mathbf{1},$$

and hence $A_t \odot V_{t+1} : X_t \nrightarrow \mathbf{1}$.

Thus $\Phi_t(x) = 1$ means that x is feasible at time t, whereas $V_t(x) = 1$ means that x is feasible at time t and that there exists a sequence of future reconfiguration actions maintaining feasibility until time N. By backward induction,

$$V_0(x_0) = \mathrm{Feas}_N(x_0).$$

2. Maximum feasible horizon without reconfiguration

At time $t = 0$, the unique Pareto-optimal convenient constellation found previously is

$$C^* = \{\{(L1, \text{SmallSat-B}, \text{SAR}), (L2, \text{SmallSat-B}, \text{SAR}), (L3, \text{SmallSat-A}, \text{Thermal})\}\}.$$

Assume that all three satellites have age 0 initially. Let $x_0 \in X_0$ be the state corresponding to C^*, with $L4$ empty. Under the no-reconfiguration policy, the action profile at every transition time $0 \le t < N$ is fixed by

$$u_t(L1) = u_t(L2) = u_t(L3) = \text{keep}, \qquad u_t(L4) = \text{idle}.$$

Under the timing convention from the previous subsection, the resulting state at time t is

$$x_t(\ell) = \begin{cases} (\text{SmallSat-B}, \text{SAR}, t), & \ell = L1, \\ (\text{SmallSat-B}, \text{SAR}, t), & \ell = L2, \\ (\text{SmallSat-A}, \text{Thermal}, t), & \ell = L3, \\ \bot, & \ell = L4. \end{cases}$$

Assuming that no additional global operational constraint forbids keeping an active satellite or leaving an empty slot idle, these profiles satisfy

$$\Gamma_t((x_t, u_t), x_{t+1}) = 1 \qquad \text{for every } 0 \le t < N.$$

Define the no-reconfiguration feasibility horizon by

$$T_{\max} = \max\{T \in \{0, 1, \ldots, N\} \mid \Phi_t(x_t) = 1 \text{ for every } 0 \le t \le T\}.$$

The altitude, radiation-tolerance, and downlink conditions do not change with time. The total science value remains

$$10 + 10 + 4 = 24 \ge 18,$$

and the occupied slots $L1, L2, L3$ continue to share the launch window Q_1. It therefore remains only to check the time-dependent power and debris-risk constraints.

Power degradation

Each SAR payload is supported by a SmallSat-B bus. Since its age at time t is t, its effective power is

$$P_B^{\text{eff}}(t) = 200(0.9)^t.$$

The SAR payload requires 150 W, so power feasibility requires

$$150 \leq 200(0.9)^t.$$

Since $0 < 0.9 < 1$, this is equivalent to

$$t \leq \frac{\log(150/200)}{\log(0.9)} \approx 2.73.$$

Thus the two SAR satellites remain power-feasible precisely for the integer times

$$t = 0, 1, 2.$$

The Thermal payload is supported by a SmallSat-A bus, so its power condition is

$$25 \leq 120(0.9)^t.$$

Equivalently,

$$t \leq \frac{\log(25/120)}{\log(0.9)} \approx 14.89.$$

Hence the Thermal satellite remains power-feasible precisely for $t \leq 14$.

Therefore, if debris risk were ignored, the first power failure would occur at $t = 3$, when the SmallSat-B buses can no longer support the SAR payloads.

Debris-risk degradation

The initial total debris risk of the occupied slots is

$$R_{L1}(0) + R_{L2}(0) + R_{L3}(0) = 0.003 + 0.001 + 0.008 = 0.012.$$

At time t, each of the three occupied slots has accumulated an additional risk of $0.001t$. Therefore

$$\begin{aligned} R_{\text{tot}}(t) &= R_{L1}(t) + R_{L2}(t) + R_{L3}(t) \\ &= \big(0.003 + 0.001t\big) + \big(0.001 + 0.001t\big) + \big(0.008 + 0.001t\big) \\ &= 0.012 + 0.003t. \end{aligned}$$

The debris-risk condition in Φ_t requires

$$0.012 + 0.003t \leq 0.012,$$

which holds if and only if

$$t = 0.$$

Consequently,

$$\Phi_0(x_0) = 1, \qquad \Phi_t(x_t) = 0 \quad \text{for every } t \geq 1.$$

Although the keep/idle action profile defines an admissible ambient transition from x_0 to x_1, its endpoint x_1 is infeasible. Equivalently, there is no corresponding arrow

$$x_0 \longrightarrow x_1$$

in the feasible transition relation

$$A_0^{\text{feas}} : X_0^{\text{feas}} \nrightarrow X_1^{\text{feas}}.$$

Hence the largest horizon on which the original Pareto-optimal constellation remains feasible without reconfiguration is $T_{\max} = 0$.

Here $T_{\max} = 0$ means that the feasible horizon consists only of the initial observation time $t = 0$.

In fact, this obstruction cannot be removed by reconfiguration. At $t = 1$, the only three-slot sets having a common launch window are $\{L1, L2, L3\}$ and $\{L1, L3, L4\}$. Their total debris risks are, respectively,

$$0.012 + 3(0.001) = 0.015$$

$$0.013 + 3(0.001) = 0.016,$$

both of which exceed 0.012. The other two three-slot sets have no common launch window. Therefore

$$\Phi_1(x) = 0 \qquad \text{for every } x \in X_1,$$

so $X_1^{\text{feas}} = \varnothing$. Thus no reconfiguration strategy can maintain convenience beyond $t = 0$ under the chosen time-dependent feasibility model.

3. Optimizing over reconfiguration strategies

The Boolean model records only whether a transition or trajectory is feasible. To compare feasible strategies by an additive numerical cost, a natural categorical choice is the Lawvere cost quantale

$$\mathsf{Cost} = ([0, \infty], \geq, +, 0).$$

The order is reversed because a smaller numerical cost is better. In this quantale, joins are ordinary infima, the monoidal product is addition, and the bottom element is ∞. This is the cost enrichment used in quantale-enriched co-design. See [2].

Equip each feasible state set X_t^{feas} with the discrete Cost-enriched category structure: the hom-value is 0 from a state to itself and ∞ between distinct states. Let

$$c_t(x, u, x') \in [0, \infty]$$

be the cost of applying action profile $u \in U_t$ to state x and obtaining x'. This may include, for example, launch, replacement, and decommissioning costs.

Define the one-step cost profunctor $d_t : X_t^{\text{feas}} \nrightarrow X_{t+1}^{\text{feas}}$ by

$$d_t(x, x') = \inf_{\substack{u \in U_t \\ \Gamma_t((x,u),x')=1}} c_t(x, u, x'),$$

with the infimum of the empty set understood to be ∞. Thus $d_t(x, x')$ is the infimal cost of an admissible action taking the feasible state x to the feasible state x', and it is ∞ when no such action exists. Equivalently, one may define d_t on the ambient state spaces X_t and assign value ∞ whenever either endpoint fails the feasibility predicate Φ_t.

Sequential composition of Cost-enriched profunctors gives

$$(d_0; \cdots ; d_{T-1})(x_0, x_T) = \inf_{x_1, \ldots, x_{T-1}} \sum_{t=0}^{T-1} d_t(x_t, x_{t+1}).$$

Hence this composite is the infimal total cost of a feasible reconfiguration strategy taking x_0 to x_T, with value ∞ if no such strategy exists.

If the terminal state is not prescribed, the optimal value over the horizon is

$$\inf_{x_T \in X_T^{\text{feas}}} (d_0; \cdots ; d_{T-1})(x_0, x_T).$$

Defining a terminal cost J_T and working backwards gives the Bellman recursion

$$J_t(x) = \inf_{x' \in X_{t+1}^{\text{feas}}} \big(d_t(x, x') + J_{t+1}(x')\big).$$

Thus Cost-enriched profunctor composition simultaneously adds successive transition costs and optimizes over intermediate states and actions.

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